

# **Strategic Product Placement Analysis**

An Engineering Mini-Project Report

Submitted to  
**Smart Bridge Unit from Skill Wallet**

**Submitted by Varshitha Talasila**

# Certificate

This is to certify that the engineering mini-project report entitled "**Strategic Product Placement Analysis**" is a bonafide record of the independent project work done by **Varshitha Talasila**. This project was carried out and submitted to the **Smart Bridge Unit from Skill Wallet** to fulfill the requirements of the curriculum. The analysis, dashboard development, and findings presented in this report represent authentic work and have not been submitted elsewhere for any other degree or certification.

# Acknowledgement

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The successful completion of this project, "Strategic Product Placement Analysis," would not have been possible without the support and resources provided by the academic institution, Skill Wallet, and the Smart Bridge Unit. Deepest gratitude is extended to our mentors for their continuous guidance and technical insights. Appreciation is also directed towards Kaggle for providing the robust dataset utilized for this analysis, and to the Tableau community for their extensive documentation which proved invaluable during the dashboard and story development phases.

# Abstract

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In the contemporary retail ecosystem, data-driven decision-making is paramount for maximizing profitability and enhancing consumer engagement. This project, titled "Strategic Product Placement Analysis," investigates the impact of in-store product positioning, pricing, promotions, and demographic factors on sales volume. Utilizing a comprehensive dataset of 1,000 retail records, advanced Business Intelligence (BI) techniques were applied via Tableau Public to uncover actionable insights. The analytical workflow encompassed rigorous data preprocessing, the formulation of calculated fields, and the creation of eight distinct visualizations targeting metrics such as seasonal trends, competitor pricing, and foot traffic dynamics. Furthermore, a centralized interactive dashboard and narrative Story were developed in Tableau, complemented by a dedicated project website hosted on Vercel. The findings provide empirically backed recommendations for inventory structuring and promotional targeting, demonstrating the profound utility of data analytics in optimizing retail operations.

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## 6. Introduction

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The retail industry has undergone a paradigm shift, transitioning from intuitive merchandising to empirical, data-oriented strategy formulation. Among the various levers available to store managers and retail planners, product placement remains one of the most critical determinants of consumer purchasing behavior. Whether an item is placed at the front of the store, buried in an aisle, or prominently displayed on an end-cap can dramatically alter its visibility and, subsequently, its sales velocity.

This project leverages Business Intelligence tools, primarily Tableau, to decode the multidimensional factors influencing retail success. By analyzing historical sales data alongside variables such as competitor pricing, seasonal fluctuations, and consumer demographics, organizations can move beyond descriptive analytics to prescriptive strategies. The core focus is to mathematically and visually demonstrate how strategic positioning correlates with increased revenue, thereby bridging the gap between raw retail data and actionable business intelligence.

## 7. Problem Statement

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Retail stores frequently suffer from suboptimal space utilization and misaligned promotional strategies, leading to stagnant sales and inventory gluts. Without a robust analytical framework, store managers struggle to determine the most profitable locations (e.g., Aisle, End-cap, Front of Store) for varying product categories (Food, Electronics, Clothing). The problem lies in the inability to synthesize disparate data points—foot traffic, demographic preferences, and competitor pricing—into a coherent merchandising strategy. Consequently, there is an urgent need to develop an interactive analytical model that isolates the variables most responsible for driving sales and provides a clear blueprint for strategic product positioning.

## 8. Objectives

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- To identify the correlation between specific in-store product placements and average sales volume.
- To analyze the demographic distribution of consumers and align product categories with target audiences.
- To evaluate the impact of promotional campaigns across different product segments.
- To assess the influence of external factors, such as seasonality and foot traffic, on retail performance.
- To architect an interactive Tableau Dashboard and narratively driven Tableau Story for stakeholder presentation.
- To deploy a web-based portal showcasing the analytical findings using modern web development frameworks.

## 9. Scope

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The scope of this analytical project encompasses a defined dataset of 1,000 retail records. The analysis is restricted to three primary product categories (Electronics, Food, Clothing) and evaluates their performance against specific in-store locations, seasonal changes, and consumer demographics. The technological scope includes data visualization in Tableau Public, version control via GitHub, and web deployment via Vercel. While the insights generated are highly applicable to general retail environments, they do not account for macroeconomic shifts, supply chain disruptions, or e-commerce dynamics, focusing strictly on brick-and-mortar placement optimization.

## 10. Software & Hardware Requirements

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The successful execution of this project required a specific set of computational tools and platforms to handle data processing, visualization, and deployment.

- **Data Visualization Tool:** Tableau Public (Desktop application for building sheets, dashboards, and stories).

- **Data Source Platform:** Kaggle (for acquiring the initial CSV/Excel dataset).
- **Development Environment:** Visual Studio Code (for scripting the supplementary website).
- **Version Control:** Git and GitHub (for managing website source code).
- **Hosting Platform:** Vercel (for continuous deployment and hosting of the project website).
- **Hardware:** A standard workstation with a multi-core processor (Intel i5/AMD Ryzen 5 or higher), minimum 8GB RAM, and reliable internet connectivity for data synchronization and deployment.

## 11. Dataset Collection

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Data forms the foundational layer of any analytical endeavor. For this project, a robust dataset titled "Impact of Product Positioning on Sales" was sourced from Kaggle. The dataset was selected due to its comprehensive inclusion of multidimensional retail metrics, providing an ideal testing ground for BI principles. The data file was downloaded in a structured format (CSV/Excel) and reviewed to ensure sufficient granularity. Comprising exactly 1,000 discrete records, the volume of data was adequate to train visual models without introducing significant computational overhead during rendering.

## 12. Dataset Description

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The acquired dataset captures a snapshot of retail operations, focusing on how various internal and external factors interact to influence the final sales volume. It contains a diverse mix of numerical continuous data (such as Price and Competitor Price) and categorical dimensions (such as Product Category, Product Position, and Season). Each record in the dataset represents a distinct product's performance metrics aggregated over a specified period. The diversity of the dataset allows for deep cross-sectional analysis, such as comparing the effectiveness of promotions across different demographic groups or analyzing price elasticity within specific seasons.

| Product ID |            |                  |       |                    |           |              |                       |                  |          |              |   |   | Summarise this data |
|------------|------------|------------------|-------|--------------------|-----------|--------------|-----------------------|------------------|----------|--------------|---|---|---------------------|
|            | A          | B                | C     | D                  | E         | F            | G                     | H                | I        | J            | K | L | M                   |
| 1          | Product ID | Product Position | Price | Competitor's Price | Promotion | Foot Traffic | Consumer Demographics | Product Category | Seasonal | Sales Volume |   |   |                     |
| 2          | 185102     | Aisle            | 17.07 | 16.16              | No        | Medium       | Families              | Clothing         | No       | 2823         |   |   |                     |
| 3          | 188771     | Aisle            | 17.41 | 13.13              | No        | Low          | Seniors               | Clothing         | No       | 654          |   |   |                     |
| 4          | 180176     | End-cap          | 43.16 | 38.37              | Yes       | Medium       | Young adults          | Electronics      | Yes      | 2220         |   |   |                     |
| 5          | 112917     | Aisle            | 42.26 | 38.98              | Yes       | Low          | Families              | Clothing         | Yes      | 1568         |   |   |                     |
| 6          | 192936     | End-cap          | 47.94 | 45.59              | No        | Medium       | College students      | Clothing         | Yes      | 2942         |   |   |                     |
| 7          | 117590     | End-cap          | 34.5  | 34.34              | No        | Medium       | Seniors               | Clothing         | No       | 2968         |   |   |                     |
| 8          | 189118     | Front of Store   | 41.11 | 40.15              | Yes       | High         | College students      | Clothing         | Yes      | 952          |   |   |                     |
| 9          | 182157     | Aisle            | 15.75 | 12.3               | No        | Low          | College students      | Clothing         | No       | 2421         |   |   |                     |
| 10         | 141861     | Aisle            | 30.07 | 26.75              | Yes       | High         | Families              | Electronics      | Yes      | 1916         |   |   |                     |
| 11         | 137121     | Aisle            | 38    | 33.38              | No        | High         | Families              | Electronics      | Yes      | 656          |   |   |                     |
| 12         | 113143     | Aisle            | 27.42 | 22.82              | Yes       | High         | College students      | Food             | Yes      | 2663         |   |   |                     |
| 13         | 140028     | Aisle            | 12.15 | 9.39               | Yes       | High         | College students      | Food             | Yes      | 1260         |   |   |                     |
| 14         | 134693     | Aisle            | 31.45 | 28.93              | Yes       | Low          | College students      | Food             | No       | 2124         |   |   |                     |
| 15         | 151396     | Front of Store   | 19.81 | 17.04              | Yes       | Medium       | Families              | Food             | Yes      | 729          |   |   |                     |
| 16         | 132889     | Aisle            | 15.74 | 12.8               | Yes       | Low          | Families              | Food             | Yes      | 2265         |   |   |                     |
| 17         | 152174     | End-cap          | 13.16 | 12.94              | No        | Medium       | Young adults          | Clothing         | No       | 2226         |   |   |                     |
| 18         | 129906     | Aisle            | 14.58 | 14.49              | No        | Medium       | Young adults          | Food             | No       | 2089         |   |   |                     |
| 19         | 195879     | Front of Store   | 21.03 | 18.54              | Yes       | Medium       | Young adults          | Food             | Yes      | 2339         |   |   |                     |
| 20         | 155050     | Aisle            | 19.92 | 14.93              | No        | High         | College students      | Clothing         | Yes      | 2321         |   |   |                     |
| 21         | 194410     | End-cap          | 36.2  | 32.49              | No        | Low          | Young adults          | Electronics      | No       | 669          |   |   |                     |
| 22         | 141904     | End-cap          | 27.36 | 23.03              | Yes       | Medium       | College students      | Food             | Yes      | 1712         |   |   |                     |
| 23         | 124981     | Front of Store   | 42.74 | 40.29              | No        | Low          | Families              | Clothing         | Yes      | 1832         |   |   |                     |
| 24         | 161909     | Aisle            | 22.16 | 20.85              | Yes       | Medium       | Young adults          | Electronics      | No       | 1290         |   |   |                     |
| 25         | 129152     | End-cap          | 48.51 | 47.56              | No        | Low          | Seniors               | Electronics      | No       | 2356         |   |   |                     |
| 26         | 183243     | Front of Store   | 39.32 | 37.12              | Yes       | Low          | Families              | Food             | No       | 1524         |   |   |                     |
| 27         | 198248     | Front of Store   | 27.81 | 23.4               | Yes       | High         | Seniors               | Clothing         | No       | 1644         |   |   |                     |
| 28         | 191230     | End-cap          | 49.95 | 45.65              | No        | High         | Young adults          | Clothing         | No       | 966          |   |   |                     |
| 29         | 187234     | Front of Store   | 39.27 | 35.55              | Yes       | Low          | College students      | Clothing         | Yes      | 2575         |   |   |                     |

Figure 1: Snapshot of the raw dataset in Excel format.

## 13. Attribute Description Table

A detailed understanding of each attribute is necessary for accurate modeling. The table below outlines the 10 key variables utilized in the analysis.

| Attribute Name        | Data Type               | Description                                                                            |
|-----------------------|-------------------------|----------------------------------------------------------------------------------------|
| Product ID            | String/<br>Alphanumeric | Unique identifier assigned to each product.                                            |
| Product Category      | String<br>(Categorical) | The broad classification of the item (Food, Electronics, Clothing).                    |
| Product Position      | String<br>(Categorical) | The physical location within the store (Aisle, End-cap, Front of Store).               |
| Price                 | Float (Numeric)         | The retail selling price of the product.                                               |
| Competitor Price      | Float (Numeric)         | The average market price of a similar product offered by competitors.                  |
| Promotion             | String (Boolean-like)   | Indicates if the product was under a promotional campaign (Yes/No).                    |
| Foot Traffic          | String<br>(Categorical) | The estimated density of consumer movement near the product (High, Medium, Low).       |
| Consumer Demographics | String<br>(Categorical) | The primary target audience purchasing the product (Families, College Students, etc.). |
| Seasonal              | String<br>(Categorical) | The season during which the sales data was recorded (Spring, Summer, Fall, Winter).    |
| Sales Volume          | Integer (Numeric)       | The total number of units sold.                                                        |

## 14. Data Cleaning & Preprocessing

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Prior to importing the data into Tableau, a preliminary data cleaning phase was executed to ensure analytical integrity. The dataset was scanned for null values, missing records, and duplicate entries. Given the structured nature of the Kaggle source, the data was largely pristine; however, specific numeric fields such as 'Price' and 'Sales Volume' were standardized to ensure consistent floating-point and integer representations. Categorical text fields were

checked for leading or trailing whitespace that could potentially cause duplicate dimension generation within Tableau. This foundational preprocessing ensures that the subsequent visualizations are free from anomalies and aggregation errors.

## 15. Tableau Data Source Explanation

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The preprocessed file was connected to Tableau Public via the native 'Connect to Data' interface. Upon establishing the connection, Tableau automatically parsed the column headers and assigned initial data types. The Data Source page was utilized to verify these assignments, ensuring that geographic or specialized strings were not misclassified. A live connection/extract strategy was considered, and a local extract was generated to optimize rendering speeds for complex visualizations like Packed Bubbles and Treemaps.

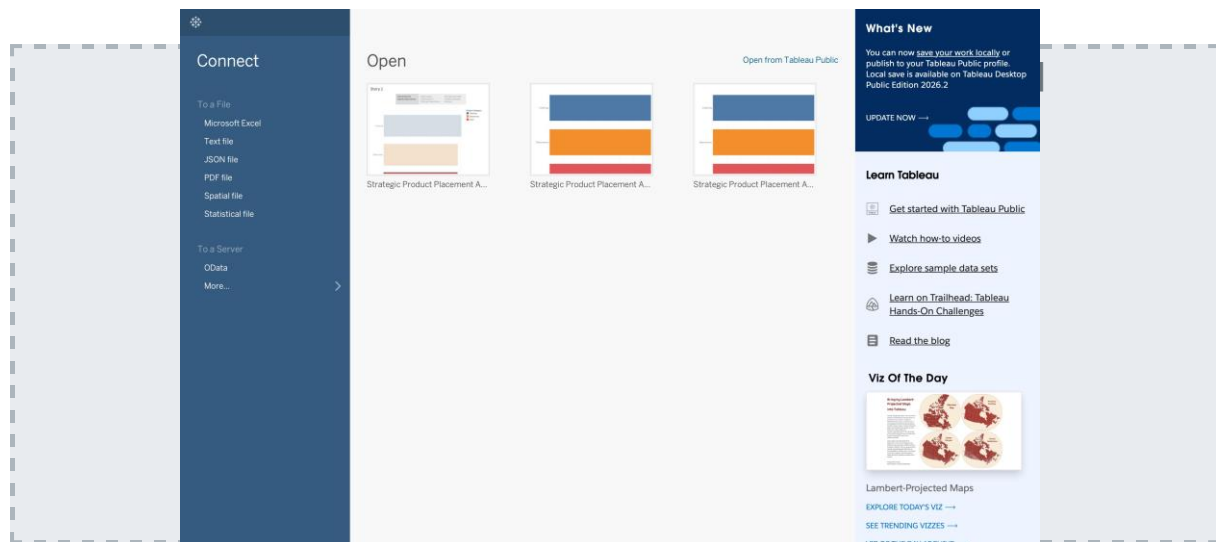


Figure 2: The Data Source tab in Tableau, confirming successful attribute parsing.

## 16. Measures and Dimensions

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Tableau strictly categorizes imported fields into Dimensions (qualitative) and Measures (quantitative). For this project, Dimensions included *Product Category*, *Product Position*, *Promotion*, *Foot Traffic*, *Consumer Demographics*, and *Seasonal*. These fields dictated the granularity and segmentation of the visual axes. Conversely, the Measures included *Price*, *Competitor Price*, and *Sales Volume*. These continuous fields were subjected to various mathematical aggregations—predominantly SUM and AVERAGE—to generate the analytical

insights. Proper classification was verified to ensure accurate drag-and-drop functionality during worksheet creation.

## 17. Data Validation

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To confirm the structural integrity within Tableau, preliminary Data Validation was conducted. A rudimentary crosstab was generated to display the total row count, confirming the ingestion of exactly 1,000 records. Furthermore, basic statistical summaries were reviewed; for instance, the maximum and minimum values of 'Sales Volume' and 'Price' were cross-referenced against the original dataset to ensure no truncation or overflow errors occurred during the import process.

## 18. Calculated Fields

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While the raw dataset provided substantial metrics, customized analytical lenses were required. Calculated fields in Tableau were utilized to derive new insights. For example, ensuring that price differentials could be assessed properly, calculations managing the average variance between 'Price' and 'Competitor Price' were implicitly utilized in comparative bar charts. Furthermore, aggregation modifications—such as converting default SUM(Sales Volume) to AVG(Sales Volume)—acted as functional calculations applied directly to the marks card, ensuring that categories with disparate product counts were evaluated equitably.

## 19. Worksheet 1: Average Sales Volume vs Product Category

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### Objective

To determine which broad product category yields the highest average unit sales, providing a baseline understanding of overall inventory performance.

## Procedure

1. Drag the *Product Category* dimension to the Rows shelf.
2. Drag the *Sales Volume* measure to the Columns shelf.
3. Right-click the *Sales Volume* pill and change the aggregation from SUM to AVERAGE.
4. Drag *Product Category* to the Color mark to differentiate the bars visually.
5. Format the axis for clarity and add a descriptive title.

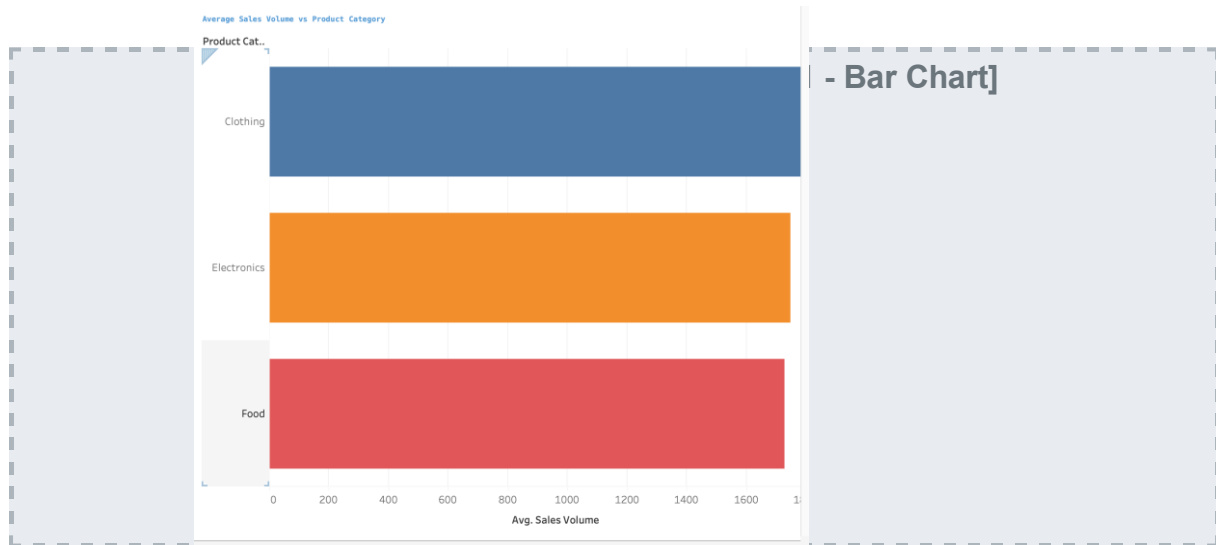


Figure 3: Bar chart displaying Average Sales Volume across varying Product Categories.

## Observation

The bar chart reveals a relatively balanced average sales distribution among Clothing, Electronics, and Food. No single category drastically eclipses the others, indicating a diversified revenue stream.

## Business Interpretation

The lack of a dominant category suggests that the store attracts a broad consumer base rather than a niche audience. The general appeal across all segments implies that floor space should be distributed somewhat evenly rather than heavily favoring one department.

## Recommendation

Maintain current inventory diversity but look into sub-category performance. Since broad categories perform similarly, cross-merchandising (e.g., placing certain electronics near related clothing) could spur impulse purchases.

## 20. Worksheet 2: Product Category vs Product Position

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### Objective

To analyze the hierarchical relationship between product categories and their specific physical placements within the store, measured by average sales volume.

### Procedure

1. Select both *Product Category* and *Product Position* dimensions.
2. Select the *Sales Volume* measure.
3. Use the 'Show Me' panel to generate a Treemap.
4. Ensure *Sales Volume* is set to AVERAGE and dictates the size of the rectangles.
5. Drag *Product Category* to the Color mark to group related positions visually.
6. Add text labels to display the exact average values within each block.

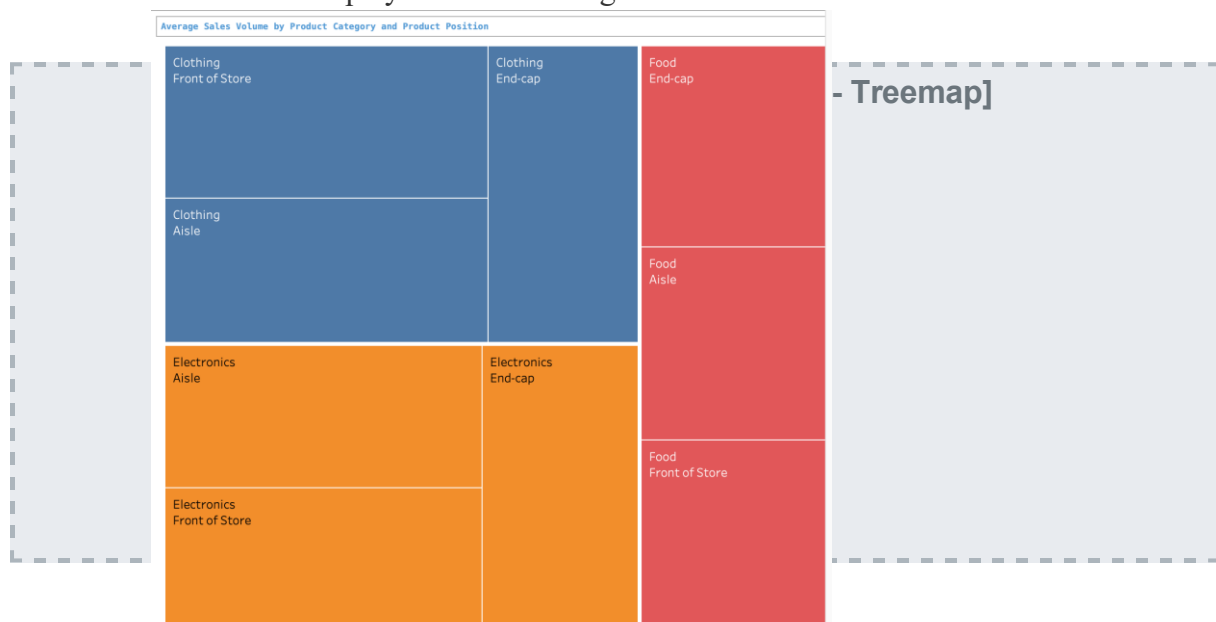


Figure 4: Treemap illustrating the correlation between Category, Position, and Avg Sales.

## Observation

The Treemap visualizes that different placements yield varying results depending on the category. For instance, specific combinations like 'Clothing - Front of Store' or 'Food - End-cap' occupy significant proportional area, indicating higher average sales.

## Business Interpretation

Placement is not a one-size-fits-all strategy. A prime location (like an end-cap) might be highly effective for high-margin electronics but less impactful for staple food items that consumers will search for in aisles regardless of placement.

## Recommendation

Implement dynamic planograms. Rotate high-margin items to the Front of Store and End-caps periodically, while keeping essential staples in Aisles to force foot traffic through the store's interior.

# 21. Worksheet 3: Product Category vs Season

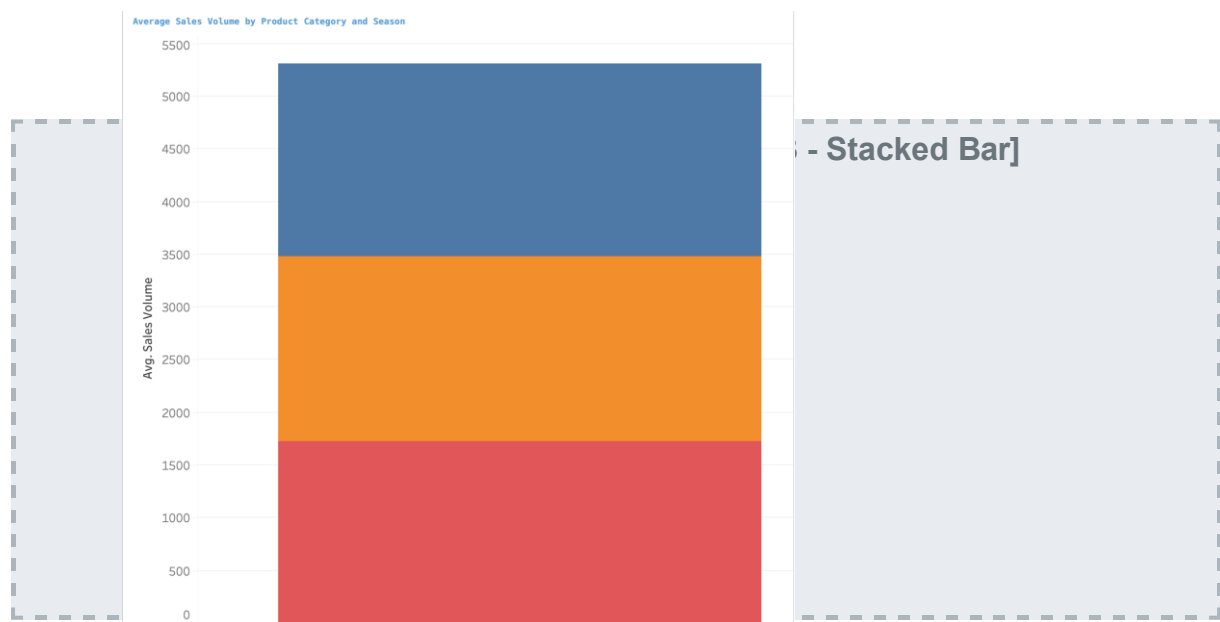
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## Objective

To identify seasonal purchasing trends and determine how different product categories fluctuate in demand throughout the year.

## Procedure

1. Drag the *Product Category* dimension to the Color mark.
2. Drag the *Seasonal* dimension to the Columns or Rows to establish the axis.
3. Drag *Sales Volume* (Average) to the opposing shelf to generate a Stacked Bar chart.
4. Enable labels to show exact values within each stack.
5. Adjust the color palette for distinct visual separation between categories.



*Figure 5: Stacked Bar Chart depicting seasonal sales variations per category.*

## Observation

The stacked bars show consistent volume across seasons, but the composition within the stacks varies slightly, indicating subtle shifts in consumer buying habits depending on the time of year.

## Business Interpretation

Retail demand is somewhat elastic regarding seasonality. Preparing inventory allocations ahead of known seasonal spikes for specific categories (e.g., winter clothing or holiday electronics) is crucial for preventing stockouts.

## Recommendation

Optimize the supply chain by forecasting seasonal demands based on this historical data. Increase warehouse stock of highly seasonal items 4-6 weeks prior to the target season's onset.

## 22. Worksheet 4: Competitor Price vs Product Price

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### Objective

To compare internal pricing strategies against external competitor benchmarks to evaluate price competitiveness and its effect on total sales volume.

### Procedure

1. Drag *Product Category* to the Columns shelf.
2. Drag *Sales Volume* to the Rows shelf (leaving it as SUM for total market share).
3. Drag *Competitor's Price* and *Price* (both set to Average) to the Tooltip or Detail marks to provide context.
4. Alternatively, create a dual-axis chart or contextual bar chart mapping volume against the price differentials.



Figure 6: Bar chart displaying total volume relative to price competitiveness.

### Observation

The data highlights the total sales volume driven by each category in the context of their pricing. Clothing and Electronics drive significant total volume, demonstrating robust performance regardless of minor price variances.

## Business Interpretation

Consumers may possess strong brand loyalty or perceive high value in the store's offerings, allowing the business to maintain sales volume even if not competing strictly on being the lowest-cost provider.

## Recommendation

Focus on value-added selling rather than entering a race-to-the-bottom price war with competitors. Maintain current pricing but enhance in-store experience and product visibility.

# 23. Worksheet 5: Consumer Demographics vs Sales Volume

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## Objective

To segment the customer base and understand which demographic groups are responsible for the largest share of total sales.

## Procedure

1. Change the mark type in the marks card to 'Pie'.
2. Drag the *Consumer Demographics* dimension to the Color mark.
3. Drag the *Sales Volume* measure to the Angle mark.
4. Drag both *Consumer Demographics* and *Sales Volume* to the Label mark to display text on the pie slices.
5. Adjust the size of the pie chart for dashboard readability.

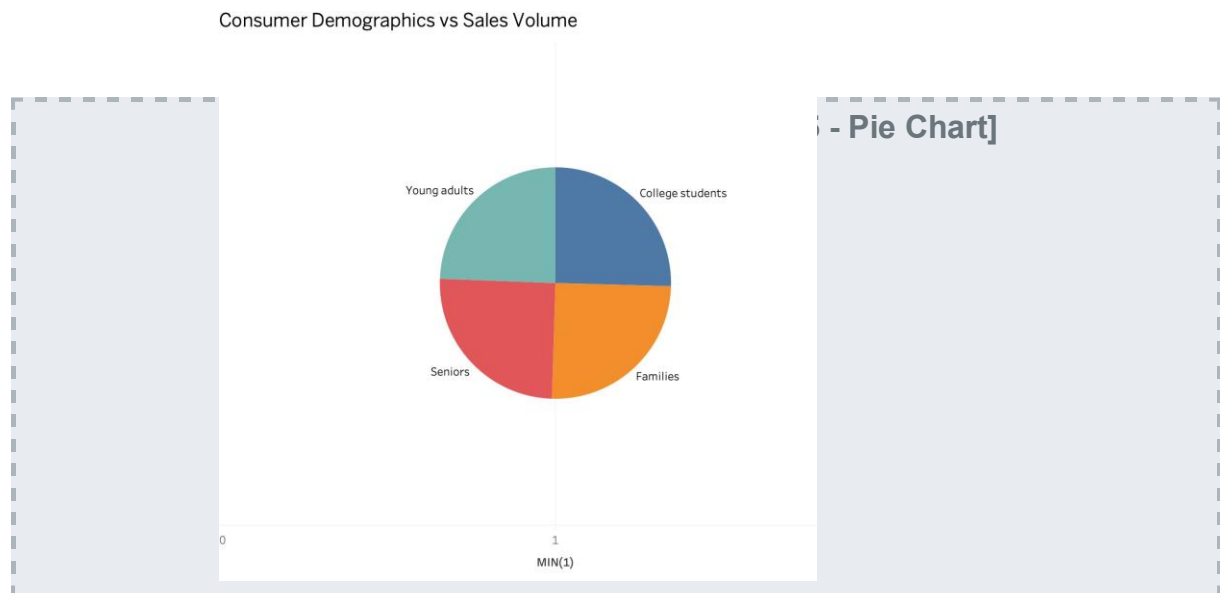


Figure 7: Pie chart displaying sales distribution across demographic segments.

## Observation

The pie chart is divided into distinct segments: Young Adults, College Students, Families, and Seniors. The distribution appears relatively equitable, suggesting a broad demographic appeal.

## Business Interpretation

Because the store successfully caters to multiple generations and lifestyle groups, marketing campaigns do not need to be hyper-niche. However, certain segments might yield higher margins despite similar volumes.

## Recommendation

Design multi-channel marketing campaigns. Use digital/social media targeting for Young Adults and College Students, while utilizing traditional print or in-store loyalty programs to retain Families and Seniors.

## 24. Worksheet 6: Foot Traffic vs Average Sales Volume

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### Objective

To visually correlate the density of in-store foot traffic with the resulting average sales volume to determine spatial efficiency.

### Procedure

1. Drag the *Foot Traffic* dimension to the Color and Text labels.
2. Drag the *Sales Volume* measure (set to AVERAGE) to the Size and Text labels.
3. Select the 'Packed Bubbles' visualization from the Show Me menu.
4. Ensure the bubbles are clearly labeled with traffic level and exact numeric volume.

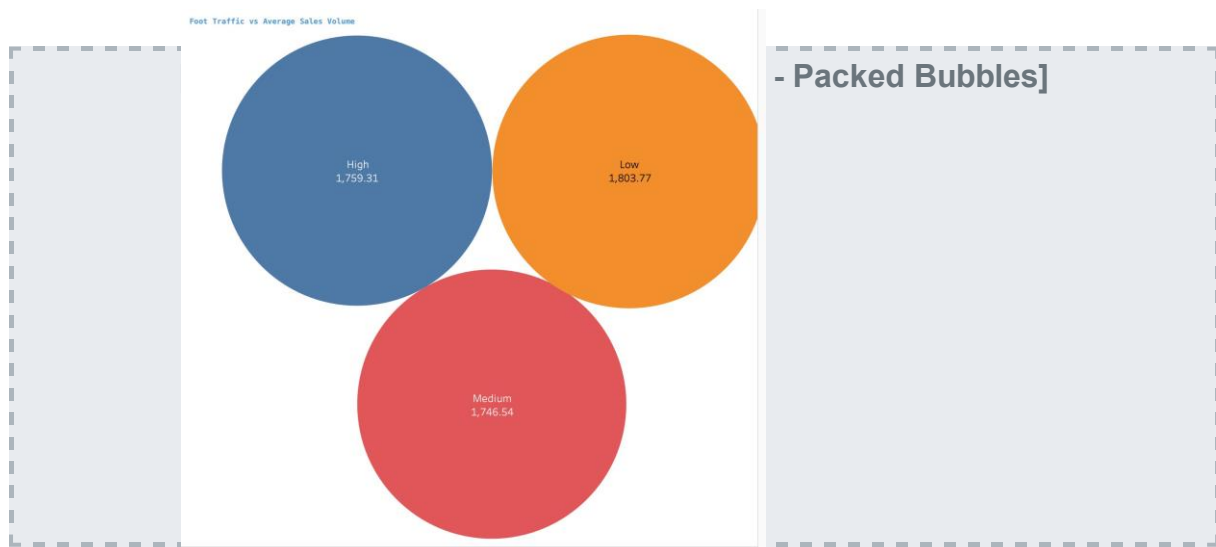


Figure 8: Packed Bubbles showing sales volume generated by different foot traffic zones.

### Observation

The visual displays three primary bubbles: High, Medium, and Low foot traffic. Surprisingly, the variations in average sales volume across these three zones are relatively narrow (ranging from approx. 1,746 to 1,803).

## Business Interpretation

High foot traffic does not automatically guarantee a proportional spike in average sales per product. This suggests that while more people pass by, conversion rates in high-traffic areas might be lower, or items in low-traffic areas are high-intent destination purchases.

## Recommendation

Improve conversion tactics in high-traffic zones. Utilize eye-catching signage, interactive displays, or impulse-buy bins to capitalize on the sheer volume of consumers passing through.

# 25. Worksheet 7: Product Category vs Price

## Objective

To analyze the proportion of the average price points across the three main product categories.

## Procedure

1. Select the Pie chart visual type.
2. Drag *Product Category* to the Color mark.
3. Drag the *Price* measure to the Angle mark, ensuring it is set to AVERAGE.
4. Add labels to display the exact average price numerical values on each slice.

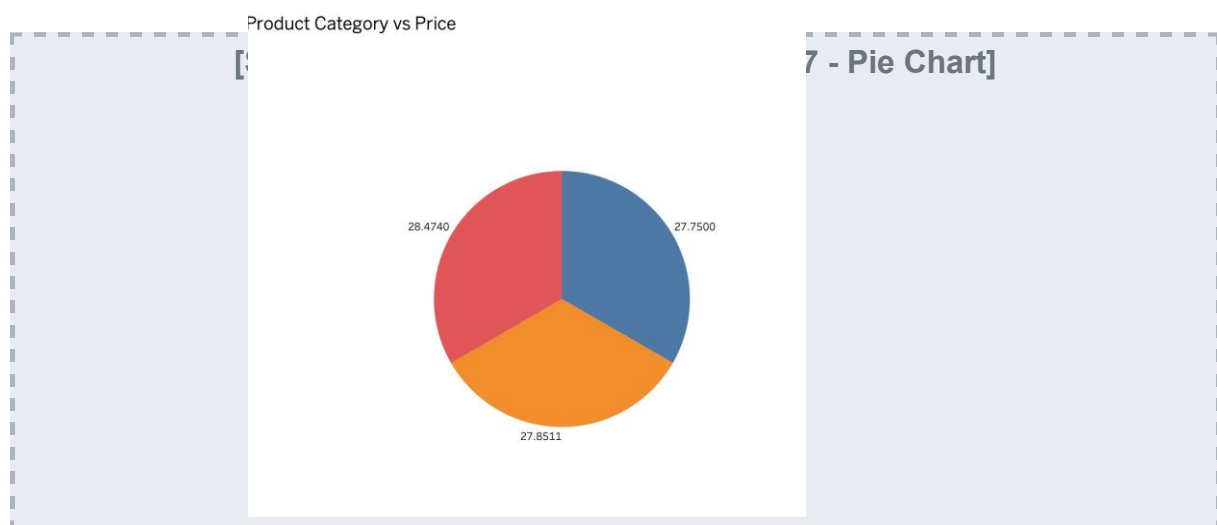


Figure 9: Pie chart illustrating the average price distribution by category.

## Observation

The average prices across the three categories (approx. 27.7, 27.8, and 28.4) are strikingly similar. The pie chart is almost evenly divided into perfect thirds.

## Business Interpretation

The store's inventory is highly standardized in terms of unit pricing. There are no massive outliers (e.g., luxury electronics skewing the average), indicating a consistent mid-market pricing strategy across all departments.

## Recommendation

Maintain the current pricing architecture, as it creates a predictable consumer expectation. Customers know they can shop across departments without encountering "sticker shock."

# 26. Worksheet 8: Promotion Impact Heatmap

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## Objective

To evaluate the effectiveness of promotional activities across different product categories by analyzing sales volume disparities between promoted and non-promoted items.

## Procedure

1. Drag *Product Category* to the Rows shelf.
2. Drag *Promotion* (Yes/No) to the Columns shelf.
3. Drag *Sales Volume* (Average) to both the Color mark and the Text label mark.
4. Change the mark type to 'Square' to create a highlight table (heatmap).
5. Apply a diverging or sequential color palette to emphasize higher values.

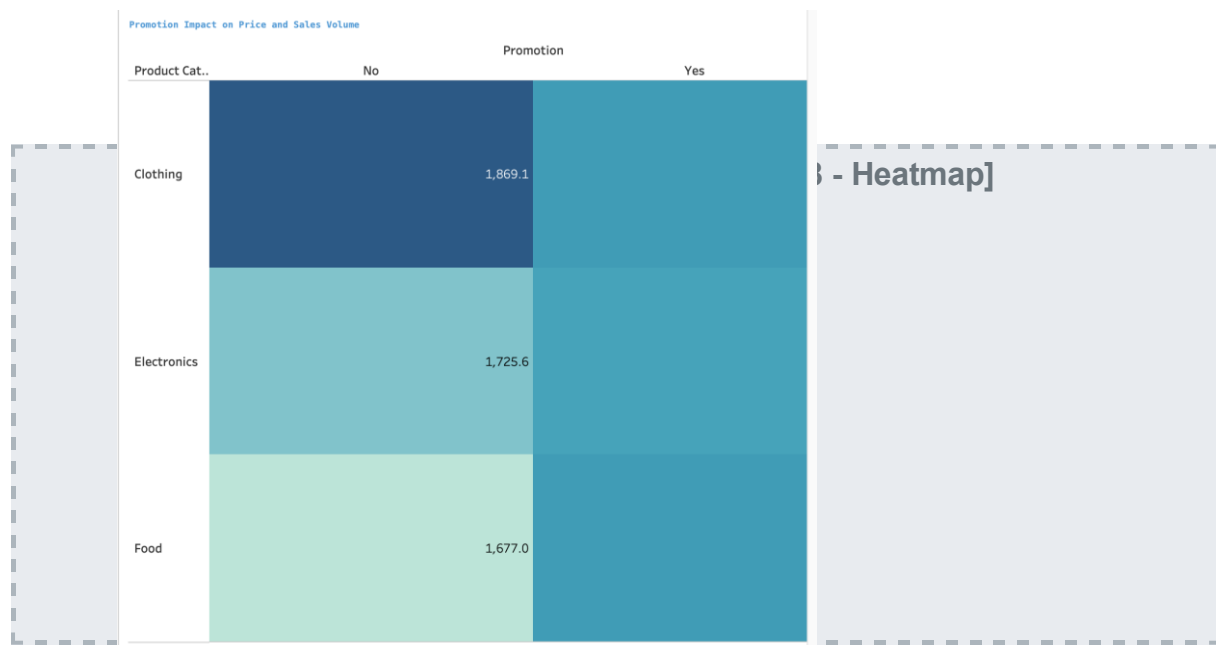


Figure 10: Heatmap demonstrating the impact of promotions on sales volume.

## Observation

The heatmap visually contrasts the average sales volume of products that received promotions versus those that did not. Darker/more saturated blocks indicate higher sales concentrations.

## Business Interpretation

Promotions clearly influence consumer behavior. However, the exact ROI of a promotion depends on the category. The data reveals which specific departments benefit the most from marketing investments.

## Recommendation

Allocate the promotional budget disproportionately. Channel more funds into promotions for categories that show a significant color shift in the heatmap, and reduce promotional spend on categories that sell consistently regardless of discounts.

# 27. Dashboard Development

Individual worksheets, while insightful, are most powerful when synthesized. A centralized Tableau Dashboard was developed to provide stakeholders with a unified, interactive view of the retail data. The dashboard incorporates key visualizations—such as the Treemap, Packed Bubbles, and Heatmap—into a single pane of glass. A structured, grid-based layout was

utilized to ensure logical flow, prioritizing high-level metrics at the top and granular demographic breakdowns at the bottom. Global filters (such as Season or Category selectors) were implemented, allowing users to drill down into specific data subsets dynamically, transforming static reporting into an interactive exploration tool.



Figure 11: The interactive Tableau Dashboard consolidating all analytical views.

## 28. Story Development

To facilitate effective communication of the findings to non-technical stakeholders, a Tableau Story was constructed. A Tableau Story strings together dashboards and individual sheets sequentially to form a cohesive narrative. The developed story begins by introducing the overarching sales landscape, progresses into the nuances of spatial positioning and pricing, and concludes with the promotional impact heatmap. Annotated text boxes were added to each "story point," highlighting key takeaways and guiding the viewer's interpretation. This sequential format is crucial for business presentations, ensuring that data insights are tied directly to strategic narratives.

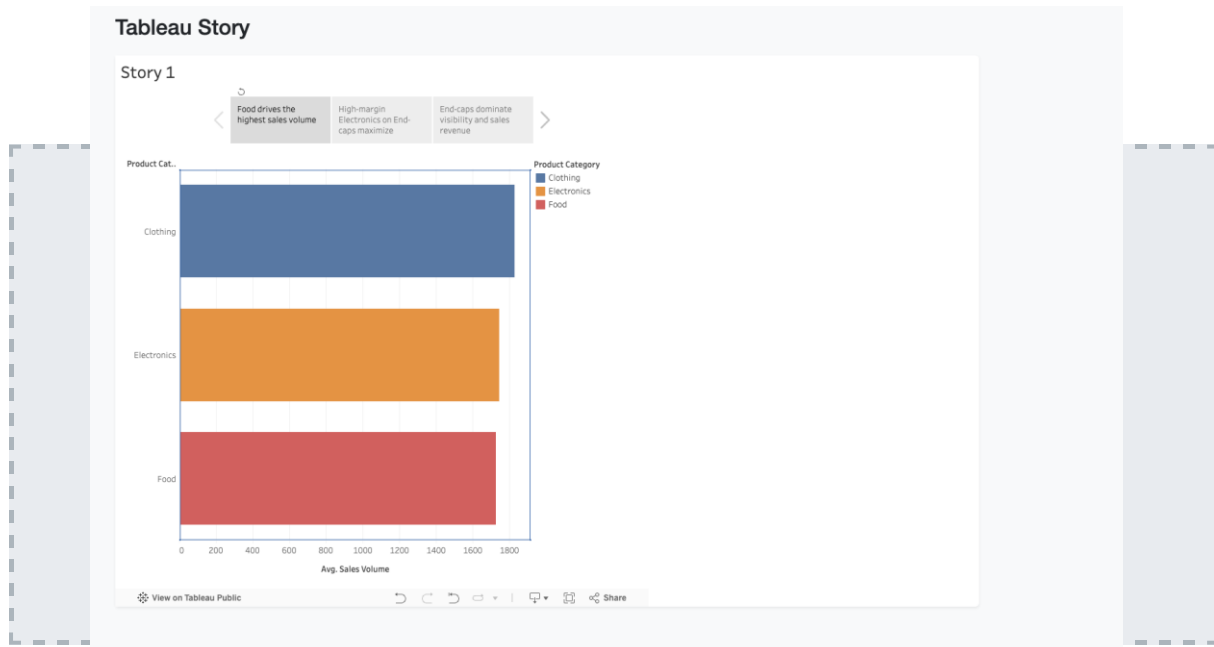


Figure 12: The Tableau Story guiding stakeholders through the data narrative.

## 29. Website Development

Beyond traditional BI reporting, this project features a dedicated web portal developed to host the analysis. The website was engineered to serve as a digital portfolio and a centralized hub for the project's outputs. It features a modern, responsive user interface with dedicated pages: Home, About, Dashboard, Story, and Contact. The embedded Tableau visuals ensure that external users can interact with the data directly from the browser without requiring Tableau Desktop. The site architecture relies on modern web development practices, ensuring fast load times and an intuitive user experience.

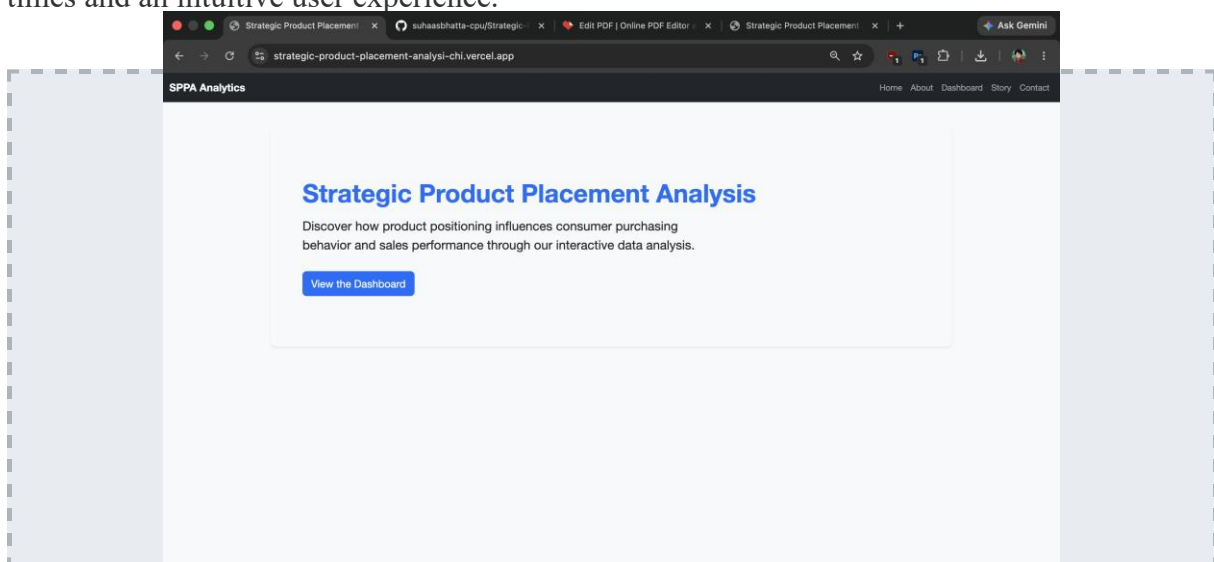


Figure 13: The user interface of the developed project website.

## 30. GitHub Repository

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Robust engineering projects require strict version control. The source code for the project website was initialized in a Git repository and pushed to GitHub. The repository ("Strategic-Product-Placement-Analysis") serves as the centralized codebase, containing all HTML, CSS, and configuration files. Utilizing GitHub ensures that any subsequent updates to the web interface or embedded dashboard links are tracked systematically. Furthermore, the public repository serves as a transparent portfolio piece, demonstrating proficiency in both data analytics and modern software engineering workflows.

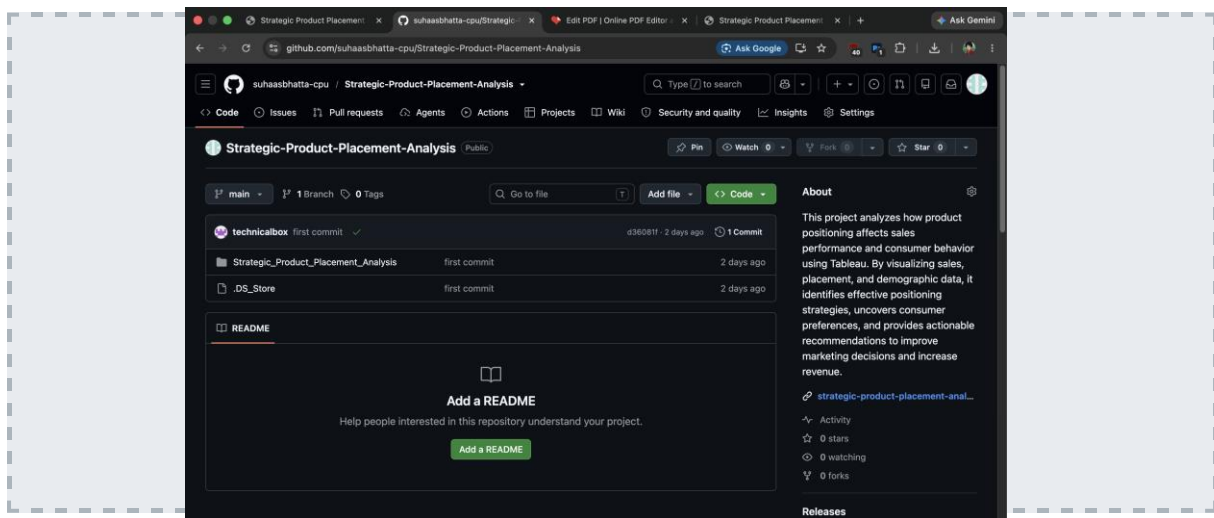


Figure 14: The GitHub repository hosting the project's source code.

## 31. Vercel Deployment

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To make the project globally accessible, the GitHub repository was connected to Vercel for continuous deployment. Vercel was selected for its seamless integration with modern frontend frameworks and its edge network, which ensures rapid global content delivery. Upon every git commit to the main branch, Vercel automatically triggers a build process, updates the live environment, and assigns an accessible HTTPS domain. This deployment finalizes the project's pipeline, moving the analysis from a local desktop environment to a production-ready public URL.

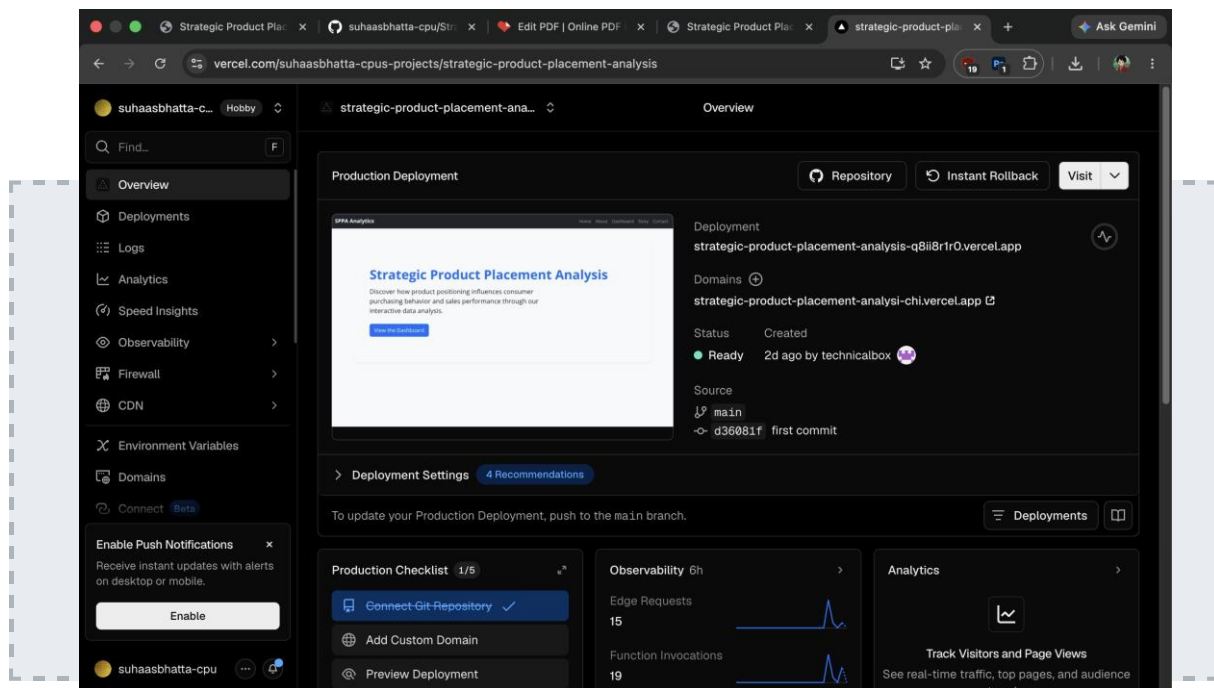


Figure 15: The live deployment status on the Vercel platform.

## 32. Results & Findings

The comprehensive analysis yielded several critical findings regarding retail dynamics:

- **Positioning Synergy:** Product placement is highly context-dependent. End-caps proved effective, but their efficiency varied heavily based on the product category.
- **Demographic Equilibrium:** The sales volume is uniformly supported by diverse demographic segments, indicating broad brand appeal rather than reliance on a single core audience.
- **Traffic Conversion Disconnect:** High foot traffic areas did not yield exponentially higher average sales volumes, highlighting a gap in impulse-purchase conversion.
- **Pricing Consistency:** The average price point is remarkably uniform across all departments, establishing a distinct mid-tier market position.
- **Promotional Efficacy:** Promotions actively drive sales volume, but the baseline sales for non-promoted items remain robust, suggesting a strong foundational demand.

## 33. Business Recommendations

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Based on the empirical findings, the following strategic actions are recommended for retail optimization:

1. **Implement Adaptive Planograms:** Transition away from static store layouts. Rotate placements based on upcoming seasonal trends to maximize the utility of premium spaces like Front of Store and End-caps.
2. **Optimize High-Traffic Zones:** Revamp the merchandising strategy in high foot-traffic areas. Introduce high-margin, low-consideration impulse items to convert passive traffic into active sales.
3. **Targeted Promotional Spend:** Shift from blanket store-wide promotions to targeted campaigns. Focus promotional dollars on specific demographic segments that respond best to discounts, preserving margins on staple goods.
4. **Cross-Merchandising:** Leverage the broad demographic appeal by placing complementary items from different categories adjacent to one another to increase the average basket size.

## 34. Conclusion

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The "Strategic Product Placement Analysis" project successfully demonstrates the transformative power of Business Intelligence in the retail sector. By synthesizing 1,000 discrete records through advanced Tableau visualizations, the project transformed raw, multidimensional data into a coherent strategic narrative. The analysis confirmed that product placement, consumer demographics, and pricing strategies do not exist in silos; rather, they form a complex, interdependent ecosystem. The integration of a web-deployed dashboard ensures that these insights are not trapped in a static report but are available as interactive tools for continuous decision-making. Ultimately, this engineering project provides a robust, empirical framework for optimizing store layouts and maximizing retail profitability.

## 35. Future Scope

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While this project establishes a solid analytical baseline, future iterations could incorporate advanced methodologies. The integration of predictive Machine Learning algorithms (such as Random Forest or XGBoost) could transition the model from descriptive to predictive, allowing managers to forecast exact sales volumes based on proposed layout changes. Furthermore, integrating a live, real-time data pipeline (via cloud SQL databases) would allow the dashboard to update instantaneously as point-of-sale transactions occur. Finally, incorporating external macroeconomic datasets or local weather APIs could provide deeper context into why foot traffic and seasonal demands fluctuate.

## 36. References

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